

Ex.no:	MINI PROJECT - 1
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Aim

To connect an Instagram Analytics dataset stored in **SQL Server** with **Power BI and Tableau**, and create interactive dashboards to analyze engagement rate, content categories, media types, and other social media performance metrics.

Algorithm

1. Create a database in SQL Server.
2. Create the InstagramAnalytics table with the required fields.
3. Import the Instagram Analytics dataset into SQL Server.
4. Verify the data using SQL queries.
5. Connect Power BI to SQL Server using **Import mode**.
6. Load the InstagramAnalytics table into Power BI and create interactive visuals using engagement rate and content category.
7. Connect Tableau to the same SQL Server database.
8. Create charts such as engagement rate by content category and posts by media type.
9. Add filters/interactivity to the dashboards.
10. Analyze the resulting visualizations.

Procedure

1. SQL Server

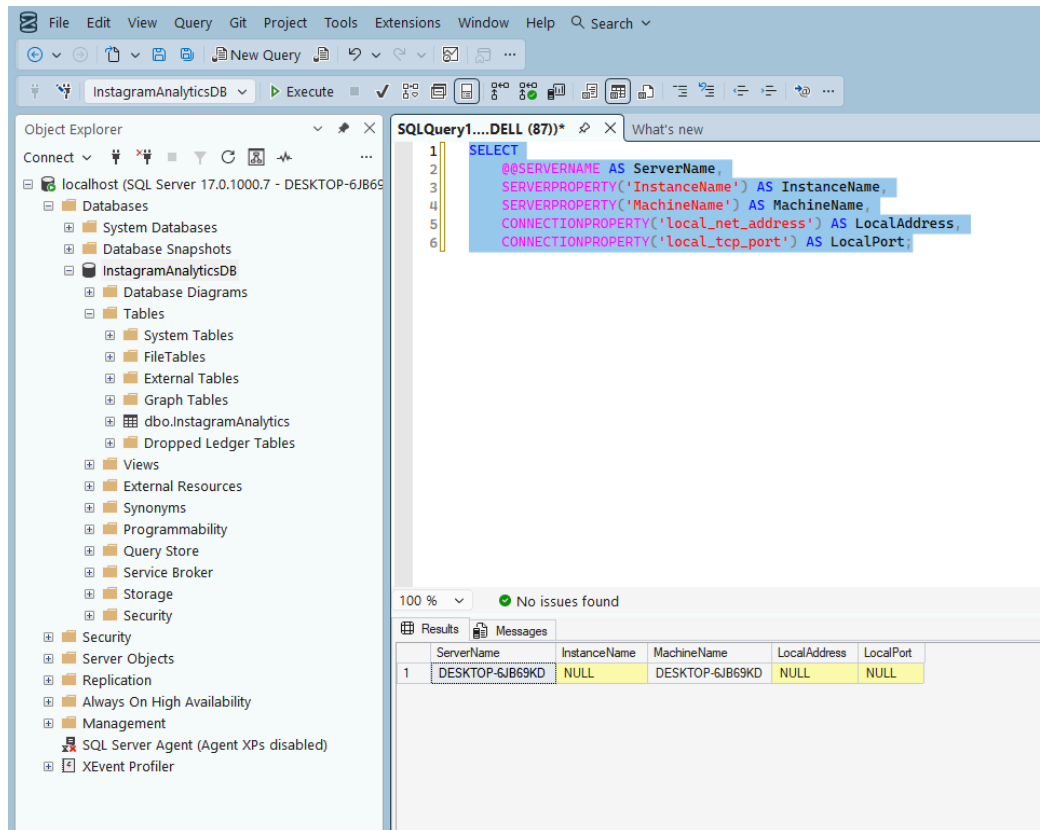
- Installed and configured **SQL Server Developer**.
- Created the database:

InstagramAnalyticsDB

- Created the table:

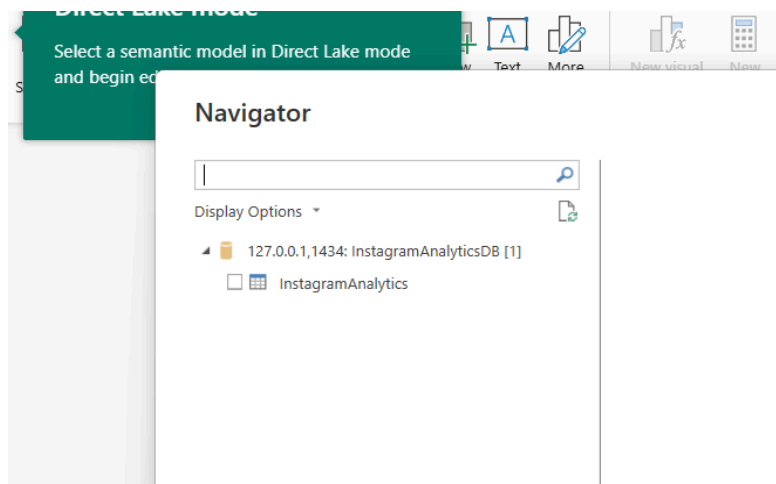
dbo.InstagramAnalytics

- Imported the cleaned Instagram Analytics dataset containing **29,999 records**.
- Verified the table and data using SQL queries.

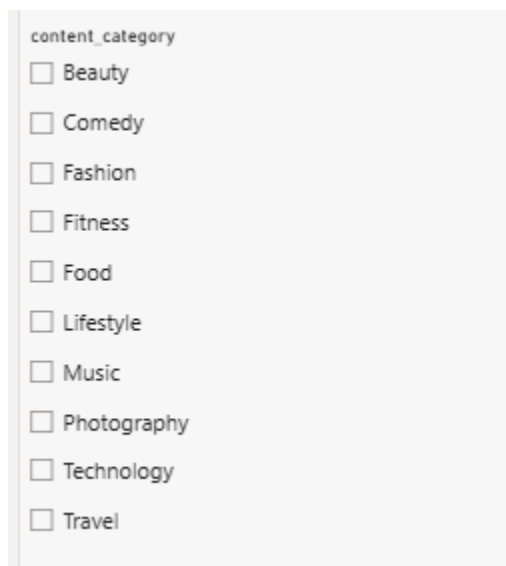


2. Power BI

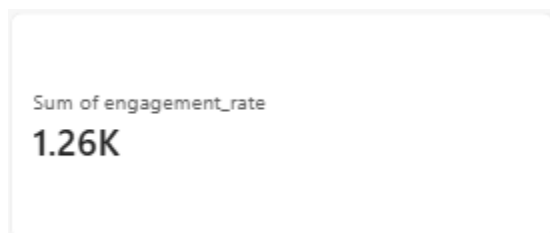
- Opened **Power BI Desktop**.
- Selected **Get Data** → **SQL Server**.
- Connected using: Server: 127.0.0.1, 1433
Database: InstagramAnalyticsDB
- Selected **Import** mode.
- Loaded the InstagramAnalytics table.
- Created an interactive dashboard containing:
 - Content Category slicer
 - Average Engagement Rate
 - Engagement Rate by Content Category
 - Posts by Media Type
- The slicer was used to interactively filter the dashboard.



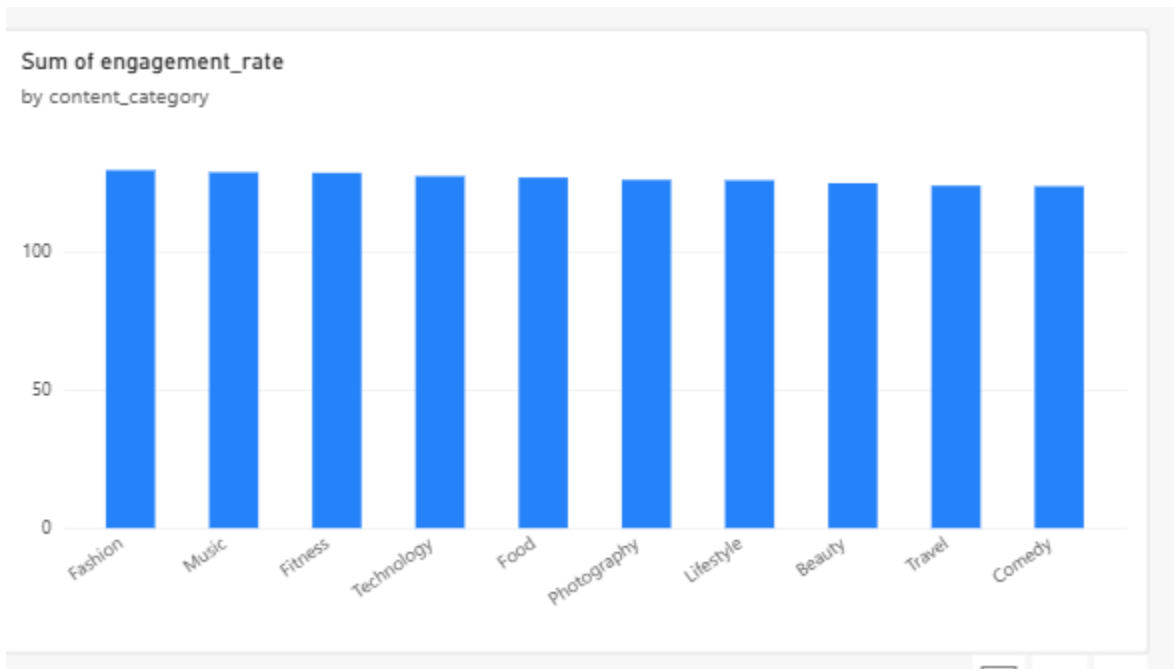
The Content Category slicer allows the user to select a specific category such as Beauty, Comedy, Fashion, Fitness, Food, Lifestyle, Music, Photography, Technology, or Travel. Selecting a category filters the other visuals in the dashboard.



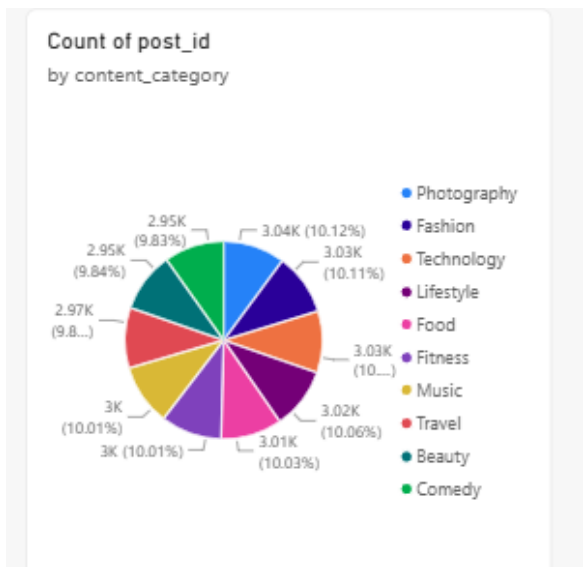
The card displays the overall engagement rate metric for the selected data. It provides a quick summary of Instagram engagement performance.



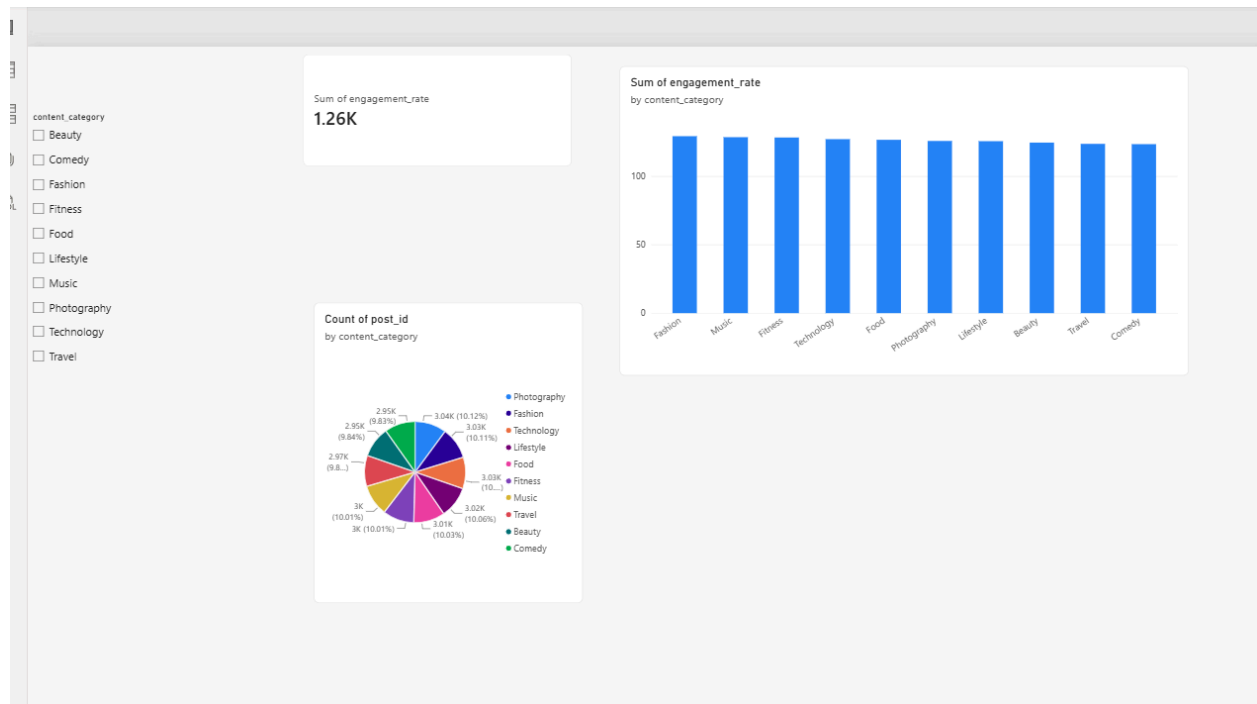
This column chart compares the average engagement rate across different content categories. It helps identify which categories generate relatively higher or lower engagement.



The pie chart represents the distribution of Instagram posts across different content categories. Each slice represents a category, while the size of the slice indicates the number and percentage of posts belonging to that category.



Final Dashboard :



3. Tableau

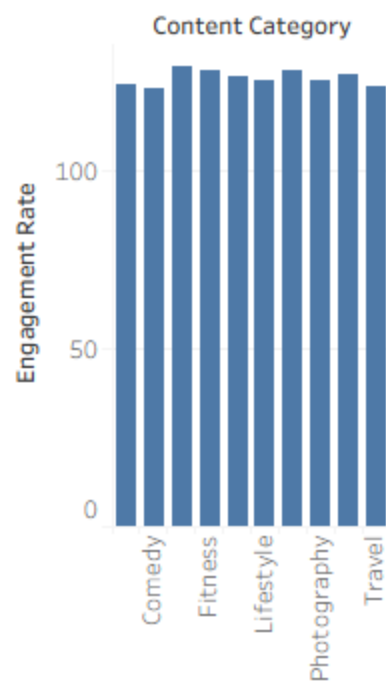
- Opened **Tableau Desktop**.
- Connected to the same SQL Server:

127.0.0.1,1433

- Selected InstagramAnalyticsDB.
- Used the InstagramAnalytics table.
- Created visualizations including:
 - Average Engagement Rate by Content Category
 - Posts by Media Type
 - Engagement Rate by Account Type
 - Reach by Traffic Source
 - Engagement Actions by Media Type
- Added filters/interactivity to analyze different content categories.

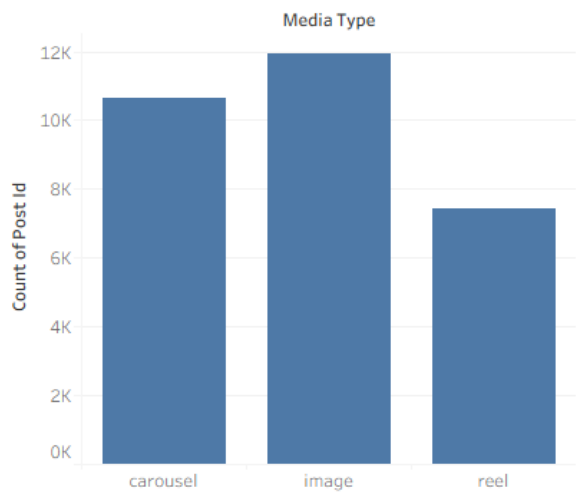
This bar chart shows the engagement rate for different content categories. It helps compare the performance of categories such as Beauty, Comedy, Fashion, Fitness, Food, Lifestyle, Music, Photography, Technology, and Travel.

Sheet 1



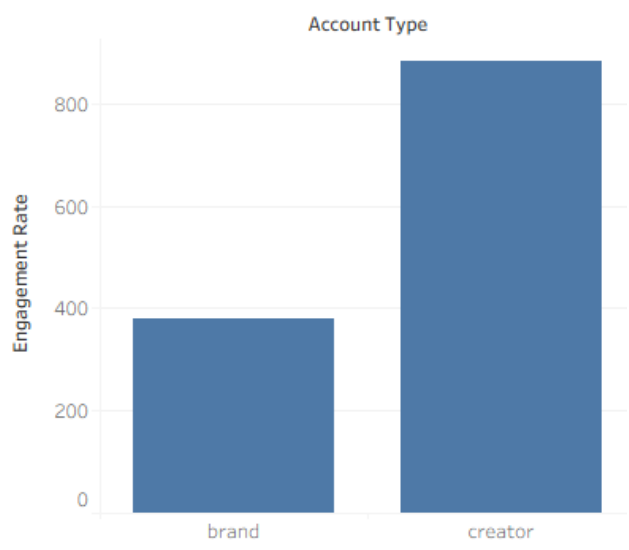
This chart represents the number of Instagram posts for each media type, including Carousel, Image, and Reel.

Sheet 2



This chart compares the engagement rate between different Instagram account types, such as Brand and Creator.

Sheet 3

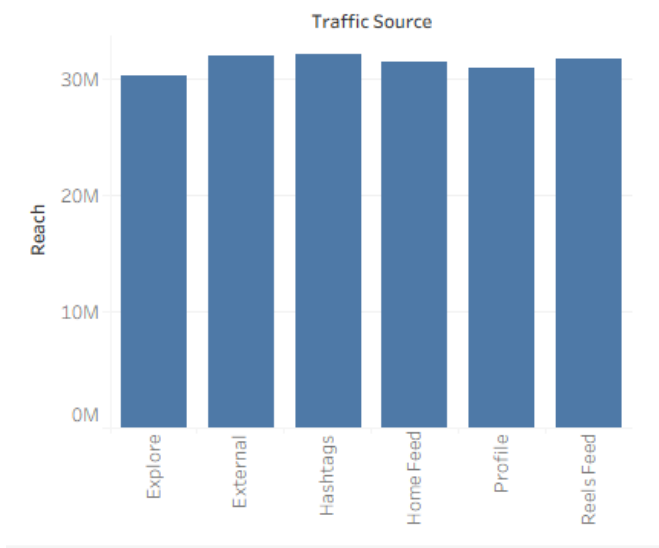


This chart compares the total reach generated through different traffic sources such as Explore, External, Hashtags, Home Feed, Profile, and Reels Feed.

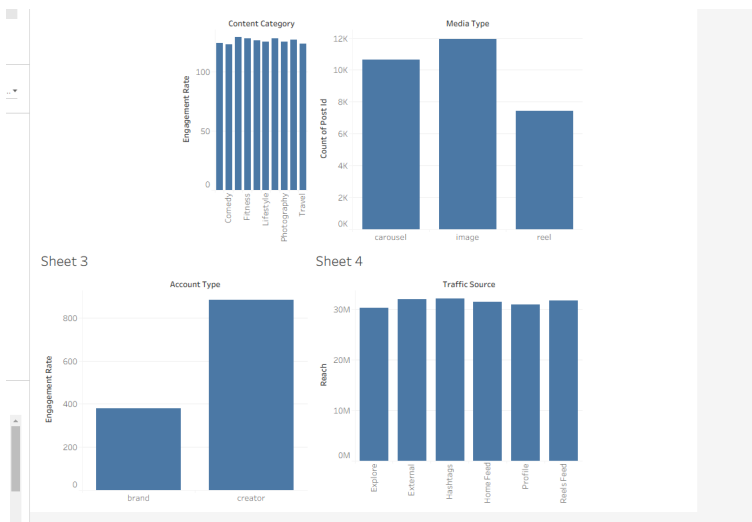
Sheet 5



Sheet 4



The Tableau dashboard combines multiple visualizations to provide an overall view of Instagram performance. It presents engagement rate by content category, post distribution by media type, engagement rate by account type, and reach by traffic source.



Result

The Instagram Analytics dataset was successfully connected to SQL Server, Power BI, and Tableau. Interactive dashboards were created to analyze social media performance based on content category,